

separate developments throughout the field. Broadly speaking, Pai predicts that 2015 will see an explosion of different materials and techniques being used in conventional 3D printing—composites, metals and new polymers. He says, “I also see low-cost SLA technology playing a larger role as prices drop, making it more affordable.” He adds, “The immediate benefit of these developments will be to product designers and developers, having the ability to prototype and test their models with a number of different materials.”

Emergence of more unusual applications of 3D printing could be expected in coming years. For instance, food industry is quickly endorsing 3D printing. Datla says, “Two kitchen-ready 3D food printers have already been presented at a trade show. These machines can print sugar, chocolate and candy in any imaginable design.”

The next phase would also be to make metal 3D printers more cost-effective. Rodagi says, “3D printing saw a boom when people had access to low-cost thermoplastic 3D printers. Now, imagine the implications of a low-cost metal 3D printer.”

Multiple-material printing would be another upcoming area. Datla says, “It will become possible to use a wider range of materials for 3D printing.” Multi-material 3D printers for various requirements of colour, elasticity and strength, and printers for electronics and optics are beginning to appear. He adds, “Combining new materials, nano-scale dimensions and printed electronics, new products with previously unimaginable properties will emerge.”

Aakash, founder and CEO, Aha3D Innovations Pvt Ltd, believes that fused filament fabrication (FFF) is firmly established as the most popular 3D-printing technology. He says, “It will continue to grow, largely because of its

Other interesting developments expected to bloom in 3D printing

- Many patents in 3D printing have already or will soon expire, which will enable even faster development and proliferation of 3D printers.
- The ability to use new printing materials, like carbon fibre and fibreglass, will make composites possible through the 3D-printing route.
- 4D printing makes it possible for objects in hard-to-reach places, such as underground water pipes, to expand or shrink depending on the water flow, thus avoiding the need to dig to exchange pipes. 4D printing involves 3D-printed items that are designed to change shape after they are produced—smart materials used to create smart structures.
- More computer graphics and CAD software will support direct printing from within the software, as easy as printing a text document with your favourite office applications.
- The possibility of printing live human organs may reduce/eliminate the requirement of donors. Customised, 3D-printed body parts will be saving peoples’ lives through artificial jawbones, splints for breathing and replacement skull parts, among a range of other uses.
- Protection of IP will be a key focus of regulations globally.
- The explosion of 3D printing from the workplace into the home will drastically cut down printing costs.

—Inputs from Vivekanandhavarma Datla,
head-Manufacturing Engineering Solutions, Cyient Ltd

Did you know?

3D printing is moving from prosthetics for humans to animals. Recently, 3D-printed prosthetics enabled a dog, born with deformed legs, to run and walk freely. Designed by 3D systems, these custom-printed prosthetics allow Derby to move faster and quicker like other dogs. 3D systems are about to print the fourth version of his prosthetics as the first version seemed too low.

Derby, the dog, with his 3D-printed prosthetics, that allow him to run freely and like any other dog



popularity and push from material innovations.”

“Currently, there is a huge gap between FFF and high-end SLS printers in print quality as well as price standard, which has brought new innovative technologies like stick deposition moulding (SDM) with anti-slip mechanism to bridge this gap,” informs Vinay Sharma, director-technical, ni logic Pvt Ltd (ni2designs). SDM gives a more accurate dose control and better precision. He adds, “A combination of such innovative improvisations in low-cost printers with advanced printing software, like Netfabb, can ensure professional-grade prints in the mid segment.”

2015 could also see more com-

mercialisation of 4D printing. This technology will entail numerous applications. Datla shares, “Examples include appliances that can adapt to heat and improve functionality or comfort, childcare products that can react to humidity or temperature, and clothing and footwear that will perform better by sensing the environment.” He adds, “4D printing makes it possible for objects in hard-to-reach places, such as underground water pipes, to expand or shrink depending on water flow, thus avoiding the need to dig to exchange pipes.”

With advancements in 3D bio-printing, the ability to synthesise and customise living tissue will likely revolutionise the field of medicine and prosthetics. However, Pai says,